

Next-Gen K-12 Learning OS

Complete Operations & Tutorial Guide

A Step-by-Step Practical Manual to Mastering the 5-Pillar Decoupled Cognitive Architecture

Author: Principal Software Engineer	Version: 2.0.0 (Production)
Target Stack: Next.js + FastAPI + PostgreSQL	Audience: Educators, Developers, Parents

Executive Overview: What Problem Does Brainoro Solve?

Traditional educational software suffers from **rigid schema constraints** (hardcoded boards and subjects) and awards opaque, unhelpful scores (like '72%') that lead to rote cramming and rapid memory decay.

Brainoro decouples cognitive mechanisms from curriculum metadata. It empowers students to achieve **measurable cognitive acceleration** across CBSE, Cambridge IGCSE, and IB MYP using 5 interconnected pillars: Universal Cornell Notes, Psychometric Adaptive Testing (IRT), Spaced Repetition (SM-2), Upward Prerequisite Remediation (DAG), and a Parent Portal tracking memory half-life.

System Map: The 5 Core Pillars

#	Pillar Name	Underlying Algorithm / Framework	User Benefit
1	Cornell Notes Layer	Universal 3-zone layout + First-Principles AI	Learn concepts intuitively in 5 mins with diagrams & worked steps.
2	Adaptive Testing	Item Response Theory (1PL/2PL Rasch Model)	Questions dynamically adjust to ability; switches CBSE/CIE/IB UX.
3	Spaced Memory Hub	SuperMemo-2 (SM-2) + 5-Box Leitner Queues	Locks formulas into long-term memory permanently without cramming.
4	Remediation Graph	Directed Acyclic Graph (DAG) Adjacency List	Traces mistakes upward to isolate exact foundational knowledge gaps.
5	Parent Acceleration Portal	Ebbinghaus Forgetting & Retention Stability S	Auditable memory half-life in days, not ambiguous percentage marks.

1. Quickstart: Launching Brainoro in 30 Seconds

Brainoro consists of a high-performance **Next.js frontend** and a decoupled **Python FastAPI cognitive backend**. Both can be started with pre-configured 1-click launcher scripts.

Step 1: Launch the Frontend Learning OS

Open Windows PowerShell or double-click the launcher at `C:\Users\pali0\Brainoro\start_frontend.bat`:

```
cd C:\Users\pali0\Brainoro\frontend npm run dev
```

Open your web browser at **`http://localhost:3000`**. The complete OS with the dynamic board switcher, Cornell notes, and test engines will load immediately.

Step 2: Launch the FastAPI Cognitive Engine

Open a second PowerShell tab or double-click `C:\Users\pali0\Brainoro\start_backend.bat`:

```
cd C:\Users\pali0\Brainoro\backend .\venv\Scripts\python.exe -m uvicorn app.main:app
--reload --port 8000
```

Interactive OpenAPI / Swagger documentation will be available at **`http://localhost:8000/docs`**.

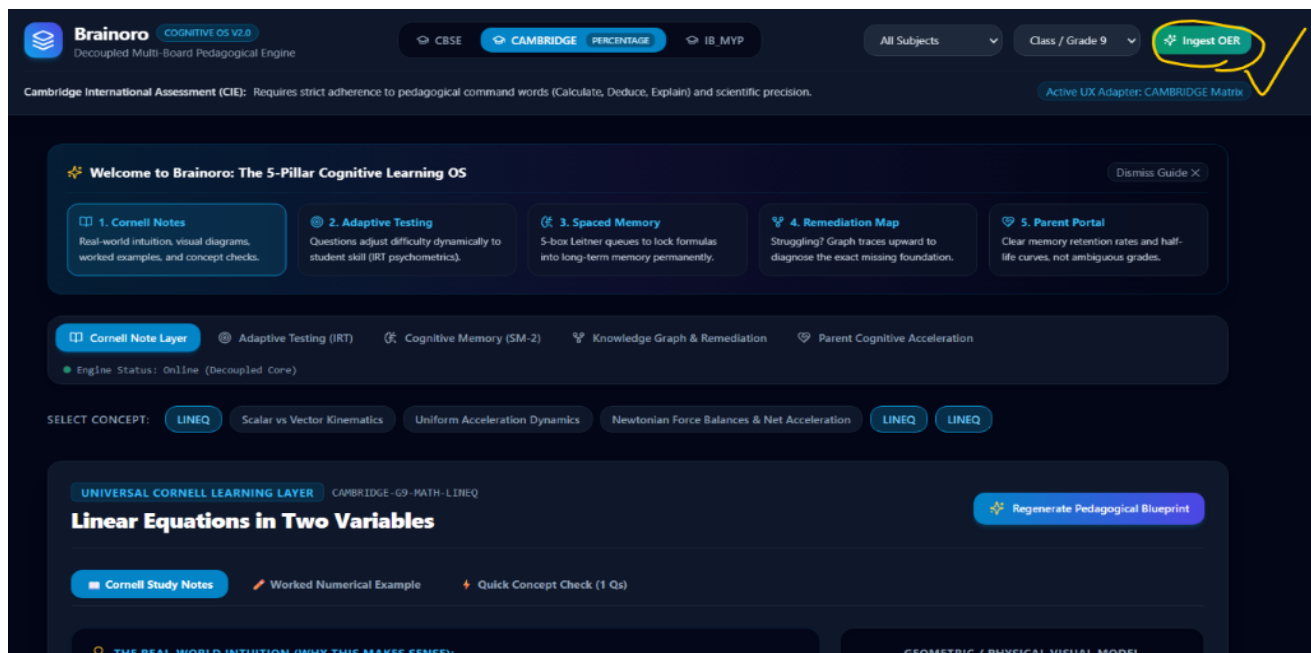


Figure 1.1: Brainoro OS Main Dashboard featuring the 5-Pillar Tour Banner, Dynamic Board Selector, and Cornell Study Layer.

2. Pillar 1: Universal Cornell Note Layer

The Cornell Note Layer transforms dense, dry textbook chapters into interactive, high-retention mental models structured in 3 synchronized learning zones.

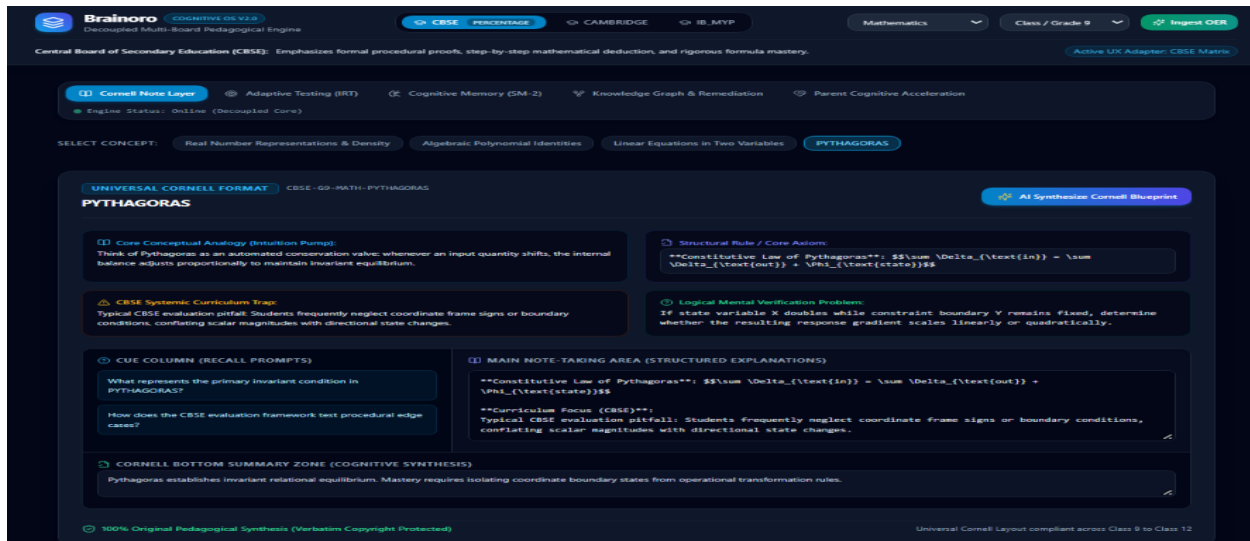


Figure 2.1: The Universal Cornell Note Structure showing Intuition Pump, Axiom Formula, Exam Traps, and Sub-Tab Navigation.

■ Quick Navigation & Topic Selection Guide:

- **Where to find Pre-Loaded Curriculums:** Look at the **Pre-Loaded Syllabus Topics** bar right below the module tabs. Brainoro comes loaded with full curriculums: **CBSE (Class 9 Math)**, **Cambridge (IGCSE Physics)**, and **IB MYP (Science)**. Simply click the board pill at the very top (CBSE / CAMBRIDGE / IB_MYP) and then click any of the numbered topic pills below!
- **Where to find '■ Quick Concept Check':** Inside the Cornell Note card, directly above the notes, click the green sub-tab **'■ Quick Concept Check (2 Qs)'** next to **'■ Cornell Study Notes'** and **'■ Worked Numerical Example'** to take an immediate interactive validation quiz.

3. Pillar 2: Dynamic Board UX & Adaptive Testing (IRT)

Unlike traditional platforms that force all students into identical multiple-choice quizzes, Brainoro dynamically changes both its **underlying psychometric difficulty** and its **board-specific user interface** at runtime.

The Psychometric IRT Engine (1PL / 2PL Rasch Model)

Every assessment item carries two calibrated parameters: **Difficulty (b)** and **Discrimination (a)**. When a student answers:

- If correct: Latent ability θ increases, and the engine queries the item bank for a more challenging question.
- If incorrect: Latent ability θ decreases, and foundational remediation is triggered.
- Next Question Selection: Uses **Fisher Information maximization** to select the exact question that provides the highest diagnostic clarity at the student's current ability boundary.

The 3 Runtime Board UX Adapters

Board Matrix	Active Features & Constraints	Pedagogical Purpose
CBSE Matrix (Class 9-10)	• Calculation Locks Active • Sequential Step-by-Step Proof Builder • Keyword validation for intermediate steps	CBSE marking schemes award marks strictly for sequential derivation. Skimming to final answers receives 0 marks.
Cambridge (CIE) (IGCSE Grade 10)	• Command Word Highlighter (<i>Calculate, Deduce, State, Explain</i>) • Interactive Scientific Calculator window • Strict 3 Significant Figures precision rule	Enforces international scientific rigor, units conformity, and official command taxonomy.
IB MYP Matrix (Middle Years 4-5)	• Criteria A–D Formative Rubrics (Scale 1-8 per criterion) • Dynamic mapping to standard 1–7 Achievement Levels • Global context inquiry prompt	Evaluates holistic inquiry, experimental design, and societal impacts rather than rote answers.

4. Pillars 3 & 4: Spaced Memory & DAG Remediation

These two mathematical engines ensure students never suffer from memory decay and can immediately diagnose why they failed an advanced problem.

Pillar 3: The SM-2 Spaced Repetition Engine

Every concept flashcard lives in an automated 5-Box Leitner queue (Box 1: Daily → Box 5: Mastered). When self-evaluating quality from 0 (Blackout) to 5 (Perfect Recall):

- **Ease Factor (EF):** Recalculates dynamically: $EF' = \max(1.3, EF + (0.1 - (5 - q) * (0.08 + (5 - q) * 0.02)))$
- **Interval Expansion:** Jumps from 1 day → 6 days → Interval * EF days on success; resets to Day 1 on failure.
- **#HardToMemorize Flag:** If tagged, penalizes interval growth by 0.75x to force higher review frequency.

Pillar 4: Knowledge Graph & Upward Remediation

Concepts are linked as a Directed Acyclic Graph (DAG). When a student fails a target concept (e.g. *Linear Equations*), the engine traverses upward along `parent_node_id` and dependency edges to identify root-cause knowledge gaps (e.g. *Real Numbers & Algebraic Identities*).

5. Content Ingestion Pipeline: Turning OER into Owned IP

Brainoro eliminates licensing fees by ingesting Creative Commons textbooks (**OpenStax** and **CK-12 FlexBooks**) through an AI transformation pipeline that strictly generates 100% original owned IP.

Step-by-Step Ingestion Tutorial:

1. Click the green '**Ingest OER**' button in the top navigation bar.
2. Select the **Target Board** (e.g. CBSE, Cambridge, or IB MYP).
3. Select the **Subject Field** (e.g. Physics or Mathematics).
4. Enter the **Topic Tracking Key** (e.g. LIGHT or PYTHAGORAS).
5. Paste the raw Markdown text from any OpenStax or CK-12 section into the text area.
6. Click '**Execute Ingestion**'.
7. The system dynamically validates the key as uppercase slug CBSE-G9-PHYSICS-LIGHT, registers the node, synthesizes the Cornell blueprint, and opens it directly in your workspace!

Open Educational Resources (OER) AI Pipeline
Ingest CK-12 FlexBooks & OpenStax text into original owned IP assets

Automated Intellectual Property Guardrails Active: The AI synthesis engine is constrained to forbid verbatim reproduction of source passages. Content is transformed into newly synthesized analogies, original step-by-step proofs, and novel validation exercises.

Target Board: CBSE Subject Field: Physics Grade Index: Class 9 Topic Tracking Key: **LIGHT**

Raw Open Educational Resource Markdown (OpenStax / CK-12):

OpenStax Algebra Section 9.3: The Quadratic Formula

For any quadratic equation in standard form $ax^2 + bx + c = 0$ where $a \neq 0$, the solutions are given by $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. The radicand $b^2 - 4ac$ is termed the discriminant and dictates whether solutions are real and distinct, real and repeated, or complex conjugates.

Successfully synthesized original concept: CBSE-G9-PHYSICS-LINEQ

Strict JSON formatting template configuration

Cancel Execute Ingestion

Figure 5.1: The OER Content Ingestion Modal binding Target Board, Subject, and Topic Tracking Key ('LIGHT') dynamically.

6. Pillar 5: The Parent Cognitive Acceleration Portal

Brainoro eliminates opaque school percentage scores (e.g. '78%') in favor of trackable, auditable memory retention metrics:

Metric	Sample Value	Practical Meaning for Parents
Retention Stability (S)	18.5 Days	The elapsed time before memory retrieval probability drops by 1/e (~37%). Indicates deep cognitive consolidation.
Memory Half-Life	12.8 Days	Exact window until there is a 50% chance of forgetting without reinforcement.
Retrieval Velocity	42 Cards/Wk	Active neural pathway reconsolidations completed each week.
Long-Term Retention	87.4%	Audited 90-day retention rate across foundational STEM concepts.

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